Aura: A Web-Based Carbon Footprint Calculator and Prediction Platform with AI-Driven Analytics and Gamification

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***Abstract*—Abstract-Most people don't know how much carbon they emit on a daily basis. For instance, going meatless once a week or turning off the lights are responsible behaviors, but they are hard to measure because there isn't enough data or consistent feedback; our team at Aura developed the web platform to specifically address this. IPCC guidelines and data from the Central Electricity Authority's Indian electricity grid are used to calculate individual carbon emissions under five categories: travel, home energy, food, general consumption, and waste.Beyond the calculation itself, Auratrack is essential because it projects future emissions based on the user's past behavior, giving users information about their current footprint rather than just where their habits are leading them. Two layers of gamification to offer incentives and feedback: Rewards are immediately transmitted to the interface following each user action. Three-tiered achievement badges are accessible through the user's past activities. "No matter what the network conditions are" is crucial. This paper describes the engineering and design decisions that resulted in the creation of Aura and makes the case that serious emission accounting, when paired with psychologically-based engagement mechanics, results in better long-term user behavior than a passive monitoring tool alone.**

Carbon footprint, sustainability, gamification, web platform, emission tracking, forecasting, Next.js, experience points, and real-time feedback are among the index terms.

***Index Terms*—carbon footprint, sustainability, gamification, web platform, emission tracking, forecasting, behavioral change, Next.js, experience points, real-time feedback**

I. Introduction

A. The Problem with Carbon Awareness Tools

First and foremost, before we even create anything, we were required to use five different carbon calculator apps.

Aura. Everyone uses it in the same way. After answering a few questions, all you get is a number. That is it. There is no more communication with you. Whether your behavior has changed is not even known to you. There is no incentive for you to use it again the next week. This is a serious problem with their design. There has been an increase in the awareness of climate change in the last 20 years. Science, however, suggests that their behavior has not changed quickly enough to stop it. This is due to a major flaw in their design. Science agrees with this assertion.

According to the IPCC's Sixth Assessment Report, "the cause of current warming, which is 1.1°C above pre-industrial levels, is cumulative emissions from household consumption patterns, which have a major influence on supply chain effects. "The difference between people's behavior and their understanding of climate change has been called the 'value action gap,' but it is not a knowledge issue. People are aware of the detrimental effects flying has on the environment. People are conscious of the carbon footprint that some foods have on the environment. People just don't have a picture of this process or anything that makes it exciting." The platform utilizes the Next.js14, TypeScript, and PostgreSQL technologies for the development of the platform, and the reward feedback to the interface occurs immediately after the user's actions despite the network conditions. Index Terms—carbon footprint, sustainability, gamification, web platform, emission tracking, forecasting, behavioral change, Next.js, experience points, real-time feedback

B. What We Built and Why

Aura was inspired by the simple observation that all the tools that successfully changed daily behaviors in other domains, such as fitness apps, language learning tools, and habit tracking tools, share one thing in common: they all have one number, show you how that number changes over time, and make improving that number feel good in the short term even if the long-term goal is still far away. Carbon tracking has none

of these things. We built Aura to fix that. Aura does three things that no other tools do: It uses regional factors instead of global averages to calculate your carbon footprint, and this makes a big difference in a country like India, where the electricity grid emission factor is significantly different from what most Western-built tools assume. It shows you where your emissions are heading based on your activity history. It gamifies all of this in a way that is informed by what behavioral psychologists say actually motivates people to stick with a task.

C. Project Objectives

Four specific goals shaped the development of this platform from the start:

1) Instead of using global data for grid intensity and transport coefficients, use data specific to the Indian context to develop an emission calculator that is sufficiently accurate for Indian users.
2) Instead of using emission forecasting that requires large amounts of historical data, use emission forecasting that requires only a small amount of historical data, since any new user will only have a few logs on day one.
3) Instead of using generic point scoring systems, develop a gamification system based on research in behavioral psychology.
4) Instead of developing a system that is only partially complete, develop it as a web application that is ready for production and has an interface that is fast enough to provide immediate feedback.

D. Paper Organization

The rest of this paper is set up like this. The Section II talks about how the work is related, Section III talks about the system architecture and how it is the most important technical choice. Section IV talks about the important parts of the implementation. In Section V, we show the results and our evaluation. In Section VI, we talk about the future scope of the project.

II. Related Work

A. Carbon Footprint Calculators

Most of them started out as spreadsheets created by environmental agencies and later evolved to the internet. Matthews et al. [2], he clearly described the problem with the accuracy of the calculations. Their study compared various tools and found that when using national average emission factors, the value was off by as much as 35 percentile from what it should have been. The problem isn't that it is hard to use, but rather that it is hard to be specific. The person who lives in Maharashtra and runs an air conditioner consumes electricity with a completely different footprint than the person who lives in Norway and runs an air conditioner. The tool that treats both of them the same way isn't really calculating their footprint. It is guessing what the footprint of the average person is and applying it to them.

Aura uses the grid emission factor that was published by India's Central Electricity Authority [3],This makes the electricity calculation much more accurate for users in India.The second limitation that we identified in the tools that are already out there was the scope of the tools.Most tools have transportation and electricity covered.Food, waste, and lifestyle consumption is either missing entirely or is available through general categorical searches with no level of activity specificity whatsoever.Aura has all five categories of activity covered, and the food emission factors are specific to dietary patterns that are typical in India.

B. Forecasting Approaches

When compared to the study of forecasting at the national level or sector level, the study of individual level emission forecasting is actually less explored.The majority of the tools for consumers do not even try. Individual level emission prediction is just one of the several areas where simple data-driven approaches can actually make a big difference without the need for large data, as discussed in the general literature on the application of computational approaches to climate change, which was reviewed in detail by Rolnick et al. [4], this helped us in personal emission prediction as one of the several areas where relatively simple and data-driven approaches helped us deliver a more meaningful impact without requiring the scale of data that more complex models need.

This directly affected our decision. Hyndman and Athanasopoulos [5] demonstrate that moving averages and linear trend methods are equivalent to complex methods when the data for training is limited for short-term forecasting. There are two months of activity logs for a new user of Aura. It would be worse than useless to train a deep learning model on it. A reasonable projection would be a least squares trend fit on the same data. impressiveness.

C. Gamification Research

The systematic review by Hamari et al. [6] The reviews were cautiously positive. The consensus on gamification results is that progress indicators, leaderboards, and points are all effective. The review also states that if gamification is implemented poorly, such as by giving points for something with no consideration for how the user might care about those points, then it has zero or very slightly negative effects. The theory for gamification comes from Self-Determination Theory. [7].The way gamification works is that if an activity satisfies all three of feeling competent, feeling autonomous, and feeling connected, then people will continue with that activity.The manner in which Aura was created was in such a way that each of these needs was being met.Competency was being met with badges from seedling to climate pion.Autonomy was being met with personal mission dashboard.Connection was being met with leaderboard.Research was done on gamification in the realm of sustainability, specifically on eco-feedback systems by Froehlich et al. [8],Their research concluded that social comparison and feedback outperformed monitoring for changing driving behavior while, although not a very large

figure, it is a real figure and can be repeated across many different study methods.

D. Interface Responsiveness

While this may seem like a trivial matter, it is actually a very important issue for the overall success of any gamification attempt. In Nielsen's seminal work on usability [9], it is stated that any response is said to be instantaneous if it is less than 100 milliseconds.

XP notification, or badgepopup, should appear as if it is responding to what you just did in the gamified app, rather than what some server somewhere in the world did a moment ago. A reward delayed is indeed a reward not received at all. The interface will immediately reflect this while the server works on it in the background asynchronously, which is why we do optimistic updates throughout the gaming layer.

III. System Architecture

A. Technology Choices

The application is a monolithic full-stack application. It is built using Next.js version 14 with the App Router. This is because the application is easier to develop and maintain as a single application. It does not have a separate frontend and backend. For a four-person project like this one, this is much easier. Also, the application uses TypeScript. This is not just for coding style. There are several logic errors in the gamification pipeline that the type checker picked up. These are errors that could have resulted in silent failures in production. Specifically, this is with regards to null values in the achievement evaluation logic. The database is a PostgreSQL database. It is hosted by Neon.tech. It uses the Prisma ORM. With Prisma, the database schema and the application code are automatically kept in sync. Fig. 1 shows the relationship of the four layers of the system.

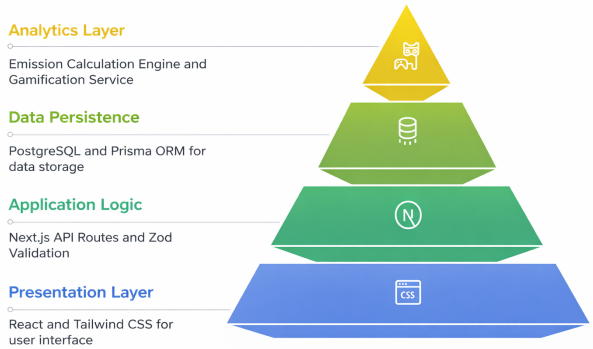


Fig. 1. Aura system architecture showing the four primary layers and their interactions.

The presentation layer deals with rendering everything using React and Tailwind. The application logic layer deals with API routes, input validation using Zod schema, and business logic. The data persistence layer deals with database read/write operations using Prisma. The analytics layer deals with a group of TypeScript service classes for calculating emissions, forecasting trends, and gamification events.

B. Security Implementation

For authentication, JSON Web Tokens are being utilized. These tokens are then stored as HttpOnly cookies. It is important to note, though, that HttpOnly is a required configuration for cookie storage, as it effectively closes one of the largest avenues for session hijacking by an attacker using XSS. Tokens are signed and expire after seven days. For each API route that handles user data, a middleware function is executed before any other code is processed.

For input validation, Zod schemas are being utilized for each API boundary. All database query code is passed through Prisma's parameterized query builder, effectively closing any avenue for a potential SQL injection attack. For rate limiting authentication routes, Upstash Redis is being utilized.

C. Emission Calculation Engine

The calculation engine converts user-logged activities into CO_2 equivalent values using category-specific emission factors. Transport emissions are calculated as:

$$E_t = D \times EF_v \tag{1}$$

where D is distance in kilometres and EF_v is the per-kilometre emission factor for the vehicle class. For electricity, the calculation uses the India grid factor of 0.82 kg CO_2e per kWh from the Central Electricity Authority [3]:

$$E_e = C \times EF_{grid} \tag{2}$$

where C is monthly consumption in kWh. The total footprint aggregates across all five categories:

$$E_{total} = E_t + E_e + E_f + E_l + E_w \tag{3}$$

Table I shows the representative emission factors the engine uses, selected specifically for relevance to Indian dietary and lifestyle patterns.

TABLE I
Emission Factors in the Aura Calculation Engine

Activity	Factor	Unit
Petrol car	0.192	kg CO_2e/km
India electricity grid	0.820	kg CO_2e/kWh
LPG cooking fuel	2.983	kg CO_2e/kg
Mutton and lamb	39.20	kg CO_2e/kg
Chicken	6.90	kg CO_2e/kg
Domestic flight	0.255	kg CO_2e/km

D. Forecasting Design

For forecasting, we purposefully avoided using a machine learning model. The rationale is pragmatic: a model without training data is just a random number generator with additional steps, and new users have no data history.Rather, a least squares linear trend is fitted to the user's monthly emission totals by the forecasting service. For a history of n months

with values $\{e_1, e_2, \ldots, e_n\}$, the projection k months ahead is:

$$\hat{e}_{n+k} = \bar{e} + \beta(n + k - \bar{t}) \tag{4}$$

where $\bar{e}$ is the historical mean, β is the OLS slope, and $\bar{t}$ is the mean time index. Projections based on national per-capita baselines are provided to users with less than three months of data, which at least provides them with something useful from their first session. In order to respond to inquiries such as "what happens to my footprint if I stop driving to work," the scenario tool allows users to modify category inputs and view updated projections in real time.

E. Gamification Architecture

Gamification System is event-driven, i.e., The system reacts appropriately when the user takes meaningful action. The Achievement Service is activated by any user action that constitutes genuine engagement. The action is compared to a table of predetermined conditions by the service.As seen in Fig. 2, if the condition is satisfied, the user receives the appropriate amount of XP, the badge is recorded, and the level is recalculated.

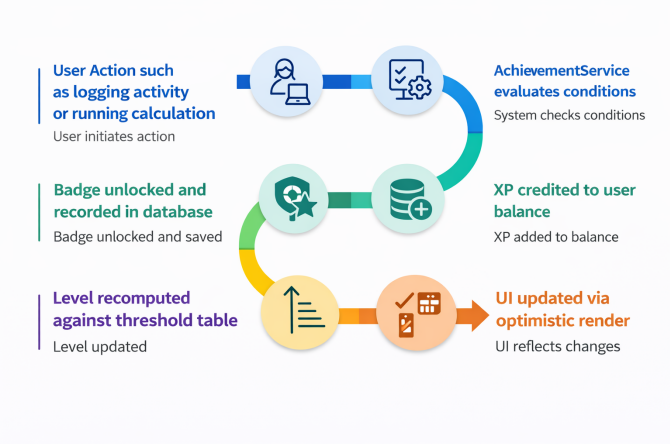


Fig. 2. Event-driven gamification pipeline from user action to interface update in Aura.

Seedling is the first level, followed by Sapling, Eco Warrior, Forest Guardian, and Climate Champion. The earlier levels appear achievable, while the later levels seem like a reward because the amount of XP increases with each level. Every time a user logs in on consecutive calendar days, the streak feature raises the count. Instead of restarting the streak, the freeze feature is designed to be triggered by one missed day per week. Internal testing revealed that streak resets caused users to instantly disengage, which prompted the implementation of this feature. This is consistent with the literature on loss aversion.

IV. Implementation Details

A. Dashboard

The dashboard is the first thing the user sees after he logs in.The dashboard shows the user the total amount of emissions by category, the recent activity feed, the current amount of XP,the current level, the current streak, and the active badge progress. Everything loads server-side on the initial request.Then subsequent updates use an optimistic UI pattern. That means that if the user logs an activity, the dashboard updates right away and the server processes the write in the background asynchronously. That makes a big difference for the user between an app that just works and an app that doesn't.

B. Achievement System

There are four levels of badges: Bronze, Silver, Gold, and Platinum.

The first badge is earned with one activity logged. This is done intentionally.

We want a user opening up our application, logging their commute, and then leaving to have something to show for it, especially if they do not come back.

Later, there are requirements such as logging seven consecutive days, completing a month with reduced emissions, or reaching certain total activity counts.

When a badge is earned, a full-screen overlay is displayed showing the name of the badge, a description of what this badge represents environmentally, and how many XP were earned.

This connects the reward system with something meaningful to the real world, which is critical to sustained engagement with such a system, according to research on eco-feedback rewards [8].

C. Leaderboard

The leaderboard displays all-time and weekly XP-related rankings.

Rankings are retrieved from a database using a direct query with server-side pagination.

We thought about calculating the rankings client-side to reduce server load, but decided against it due to scalability issues,since we would have to send the entire user list to each client. Users that are lower in rank have their own stats prominently displayed next to the ranking table.

D. Account Deletion

The whole process of deleting an account is a single transaction that cascades through all the tables that contain information about the user, from the activity log to the gamification log to the achievement log to the session token log to the user log. However, prior to making this request, a confirmation string is entered into a field provided by the interface. This is a design choice. Irrevocable actions should not be made too easily; a pause is provided in this case.

V. Results and Evaluation

A. Performance Measurements

Under normal load conditions, we have recorded API response times for our application during deployment on a local and a staging level. The main API calls for calculating carbon

and aggregating the dashboard have a consistent response time between 200 and 350 milliseconds. Activity logs have a delay of 45 milliseconds due to the gamification pipeline. However, users are unaware of this delay. For user history up to 24 months long, the forecasting service provided a forecast for six months in less than 80 milliseconds. These values are important because they confirm that our architectural decision to perform computation on the server-side is correct. We have contemplated moving the forecasting computation to the client-side in order to reduce server load; however, this computation is already sufficiently fast that such a move is not necessary.

B. Functional Testing Results

In the entire application, every important user process was manually tested. This included logging in, registering, recording activities, and calculating

In every test case, account deletion, streak accumulation, leaderboard positioning, achievement unlocking, and tion were all successfully executed with actual database persistence. In the process of testing our program, an issue came up with the achievement evaluation order in cases where several activities were recorded in quick succession after one another. The total XP calculation sometimes updated before the evaluation of the badges was complete due to sequential execution in the AchievementService. We fixed this by ensuring that the evaluation is complete before the XP increment is executed. There were no compilation errors in the final version of the program in TypeScript.

C. Comparison With Existing Tools

Table II places Aura against three publicly available carbon footprint platforms on the features most directly relevant to long-term user engagement.

TABLE II
FEATURE COMPARISON ACROSS PLATFORMS

Feature	Aura	WWF	Cool	CF.com
Multi-category	Yes	Yes	Yes	Yes
Forecasting	Yes	No	No	No
Gamification	Yes	No	No	No
What-if tool	Yes	No	Yes	No
Region factors	Yes	No	Partial	No
Real-time UI	Yes	No	No	No

D. Limitations

There are three limitations that need to be mentioned. The first one is that the user buying locally sourced vegetables gets the same factor as the user buying imported ones, as the food and lifestyle factors are based on lifecycle assessment averages, which don't account for differences in the local supply chain. Secondly, the assumption of the forecasting model doesn't hold for users whose behavior changes suddenly. Finally, as we didn't perform a user study, the literature cited only shows that gamification increases user retention in the long term, but this was not tested on the user base of Aura.

VI. CONCLUSION

A. What We Built and What It Shows

Aura is a solution to a specific question: can we build a carbon tracking system that people will use? We took the engagement problem as seriously as we took the calculation problem. When you know the right emission factors, the math of emissions is simple. The hard part is building something that is worthy of a place in someone's routine. None of these features—the leaderboard, the XP, the tiered badges, the streaks, and the gamification—were ever an afterthought.They were designed from the beginning to satisfy certain human psychological needs that, according to behavioral psychology literature, drive long-term engagement. Only a long-term user study can validate whether this architecture truly scales. However, the point of this architecture is to test it. With optimistic updates throughout the UI, the technology stack of Next.js 14, TypeScript, PostgreSQL, and Prisma delivered the responsiveness required to drive this gamification model. The user interface receives each reward notification in a timely fashion to associate it with the action taken to earn it

B. Where We Go Next

The best ways in which this work could be extended in a useful manner are in the following four areas. The biggest problem with the current system, which is that people have to log their activities by hand, would be completely solved by combining the smart meter's APIs with the car's telematics. A mobile app would send out achievement notifications and streak reminders through the channel where most users spend most of their time. A controlled longitudinal study comparing Aura users with passive carbon calculator users would yield evidence demonstrating the inadequacies of the current evaluation process. So, it is both technically possible and scientifically interesting to replace the linear trend forecasting with a custom model once the platform has enough anonymized usage data.

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